

Assignment

Date _____ Period _____

Simplify. Write your answer in rectangular form.

1) $4\left(\cos \frac{2\pi}{3} + i\sin \frac{2\pi}{3}\right) \cdot 6\left(\cos \frac{4\pi}{3} + i\sin \frac{4\pi}{3}\right)$

Product theorem: multiply the r's, add the angles.

$$r = 4 \cdot 6 = 24$$

$$\theta = \frac{2\pi}{3} + \frac{4\pi}{3} = \frac{6\pi}{3} = 2\pi$$

 2π is a full turn, so it lands on 0. Keep angles in $[0, 2\pi)$.

$$24(\cos 0 + i\sin 0) = 24(1 + 0i)$$

$$\boxed{24}$$

A product can land exactly on the real axis. $24 + 0i$ is just 24.

2) $6\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right) \cdot 4\left(\cos \frac{5\pi}{4} + i\sin \frac{5\pi}{4}\right)$

Multiply r's, add angles (common denominator 12).

$$r = 6 \cdot 4 = 24 \quad \theta = \frac{2\pi}{12} + \frac{15\pi}{12} = \frac{17\pi}{12} \quad (255^\circ)$$

 255° is not on the unit circle. Split it into two angles that are: *$\frac{17\pi}{12} = \frac{7\pi}{6} + \frac{\pi}{4}$ ($210^\circ + 45^\circ$), then use the sum formulas.*

$$\cos \frac{17\pi}{12} = \cos \frac{7\pi}{6} \cos \frac{\pi}{4} - \sin \frac{7\pi}{6} \sin \frac{\pi}{4}$$

$$= \left(-\frac{\sqrt{3}}{2}\right) \frac{\sqrt{2}}{2} - \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} = \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$\sin \frac{17\pi}{12} = \sin \frac{7\pi}{6} \cos \frac{\pi}{4} + \cos \frac{7\pi}{6} \sin \frac{\pi}{4}$$

$$= \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} + \left(-\frac{\sqrt{3}}{2}\right) \frac{\sqrt{2}}{2} = \frac{-\sqrt{2} - \sqrt{6}}{4}$$

$$24 \left(\frac{\sqrt{2} - \sqrt{6}}{4} + i \frac{-\sqrt{2} - \sqrt{6}}{4} \right)$$

$$\boxed{6\sqrt{2} - 6\sqrt{6} + (-6\sqrt{6} - 6\sqrt{2})i}$$

Faster route: each factor IS on the unit circle. $6\text{cis} \frac{\pi}{6} = 3\sqrt{3} + 3i$ and $4\text{cis} \frac{5\pi}{4} = -2\sqrt{2} - 2\sqrt{2}i$. FOIL them and you get the same answer with no sum formula. Check: 255° is QIII, both parts negative.

3) $(2 - i)(-2 + 5i)$

Rectangular times rectangular: FOIL, then replace i^2 with -1 .

$$(2 - i)(-2 + 5i) = (2)(-2) + (2)(5i) + (-i)(-2) + (-i)(5i)$$

$$= -4 + 10i + 2i - 5i^2$$

$$= -4 + 12i - 5(-1) = -4 + 12i + 5$$

$$\boxed{1 + 12i}$$

The trap is the last term: $(-i)(5i) = -5i^2 = +5$. The sign flips because $i^2 = -1$.

$$4) \frac{-5 + 4i}{4 + 5i}$$

Dividing in rectangular form: multiply top and bottom by the conjugate of the bottom, $4 - 5i$. That makes the bottom real.

$$\text{Top: } (-5 + 4i)(4 - 5i) = -20 + 25i + 16i - 20i^2 \\ = -20 + 41i + 20 = 41i$$

$$\text{Bottom: } (4 + 5i)(4 - 5i) = 16 + 25 = 41$$

$(a + bi)(a - bi) = a^2 + b^2$. Always a plus.

$$\frac{41i}{41}$$

$$\frac{41i}{41}$$

$$\boxed{i}$$

$$5) \frac{4 + 3i}{2i}$$

The bottom is pure imaginary. Multiply top and bottom by $-i$ (the conjugate of $2i$ is $-2i$) so the bottom turns real.

$$\text{Top: } (4 + 3i)(-i) = -4i - 3i^2 = -4i + 3 = 3 - 4i$$

$$\text{Bottom: } (2i)(-i) = -2i^2 = 2$$

$$\frac{3 - 4i}{2} = \frac{3}{2} - \frac{4}{2}i$$

$$\boxed{\frac{3}{2} - 2i}$$

Split into $a + bi$ and reduce each fraction.

$$6) \frac{4 - 6i}{-6 - 2i}$$

Conjugate of the bottom is $-6 + 2i$.

$$\text{Top: } (4 - 6i)(-6 + 2i) \\ = -24 + 8i + 36i - 12i^2$$

$$= -24 + 44i + 12 = -12 + 44i$$

$$\text{Bottom: } (-6)^2 + (2)^2 = 36 + 4 = 40$$

$$\frac{-12 + 44i}{40} = -\frac{12}{40} + \frac{44}{40}i$$

$$\boxed{-\frac{3}{10} + \frac{11}{10}i}$$

$(-6i)(2i) = -12i^2 = +12$. Reduce both fractions by 4.

$$7) (6(\cos 30 + i \sin 30))^4$$

De Moivre: raise r to the 4th, multiply the angle by 4. The angle is in degrees, so the answer stays in degrees.

$$6^4 = 1296 \quad 4 \cdot 30^\circ = 120^\circ$$

$$1296(\cos 120^\circ + i \sin 120^\circ)$$

$$= 1296 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$\boxed{-648 + 648\sqrt{3}i}$$

120° is QII: $\cos$ negative, $\sin$ positive. Distribute 1296 to both parts.

$$8) (5 - 3i)^4$$

$\tan \theta = -\frac{3}{5}$ is not a unit-circle value, so De Moivre can't give exact values. Square it, then square again: $(z^2)^2 = z^4$.

$$(5 - 3i)^2 = 25 - 30i + 9i^2$$

$$= 25 - 30i - 9 = 16 - 30i$$

$$(16 - 30i)^2 = 256 - 960i + 900i^2$$

$$= 256 - 960i - 900$$

$$\boxed{-644 - 960i}$$

$(a + bi)^2 = a^2 - b^2 + 2abi$. Never square each part separately.

$$9) (5 - 6i)^4$$

Same move: not a special angle, so square twice.

$$(5 - 6i)^2 = 25 - 60i + 36i^2$$

$$= 25 - 60i - 36 = -11 - 60i$$

$$(-11 - 60i)^2 = 121 + 1320i + 3600i^2$$

$$= 121 + 1320i - 3600$$

$$\boxed{-3479 + 1320i}$$

$2(-11)(-60) = +1320$: two negatives make the middle term positive.

Find all n th roots. Write your answers in polar form.

$$10) 1 + i\sqrt{3}, n = 5$$

Convert to trig form, then the n th root theorem:

$$\sqrt[n]{r} \text{cis} \left(\frac{\theta}{n} + \frac{2\pi k}{n} \right).$$

$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{\sqrt{3}}{1} = \sqrt{3}, \text{ QI, so } \theta = \frac{\pi}{3}$$

$$\sqrt[5]{2} \quad \text{first: } \frac{\pi/3}{5} = \frac{\pi}{15} \quad \text{spacing: } \frac{2\pi}{5} = \frac{6\pi}{15}$$

Common denominator 15, then keep adding $\frac{6\pi}{15}$ until you have 5 roots.

$$\sqrt[5]{2} \text{cis} \frac{\pi}{15}, \sqrt[5]{2} \text{cis} \frac{7\pi}{15}, \sqrt[5]{2} \text{cis} \frac{13\pi}{15},$$

$$\sqrt[5]{2} \text{cis} \frac{19\pi}{15}, \sqrt[5]{2} \text{cis} \frac{5\pi}{3}$$

$\frac{25\pi}{15} = \frac{5\pi}{3}$. In degrees: $12^\circ, 84^\circ, 156^\circ, 228^\circ, 300^\circ$ (same roots).

$$11) -\frac{5}{2} + \frac{5\sqrt{3}}{2}i, n = 5$$

Convert first. Both parts have a denominator of 2, so square carefully.

$$r = \sqrt{\left(-\frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} = \sqrt{\frac{25}{4} + \frac{75}{4}} = \sqrt{25} = 5$$

$$\tan \theta = \frac{5\sqrt{3}/2}{-5/2} = -\sqrt{3}, \text{ ref } 60^\circ, \text{ QII, so } \theta = 120^\circ$$

$$\sqrt[5]{5} \quad \text{first: } \frac{120^\circ}{5} = 24^\circ \quad \text{spacing: } \frac{360^\circ}{5} = 72^\circ$$

$$\sqrt[5]{5} \text{cis} 24^\circ, \sqrt[5]{5} \text{cis} 96^\circ, \sqrt[5]{5} \text{cis} 168^\circ,$$

$$\sqrt[5]{5} \text{cis} 240^\circ, \sqrt[5]{5} \text{cis} 312^\circ$$

$$\left(\frac{5\sqrt{3}}{2}\right)^2 = \frac{25 \cdot 3}{4} = \frac{75}{4}. \sqrt[5]{5} \text{ doesn't simplify, so leave it.}$$

$$12) \frac{5}{2} - \frac{5\sqrt{3}}{2}i, n = 2$$

Square roots ($n = 2$): two answers, 180° apart.

$$r = \sqrt{\left(\frac{5}{2}\right)^2 + \left(-\frac{5\sqrt{3}}{2}\right)^2} = \sqrt{\frac{25}{4} + \frac{75}{4}} = \sqrt{25} = 5$$

$$\tan \theta = \frac{-5\sqrt{3}/2}{5/2} = -\sqrt{3}, \text{ ref } 60^\circ, \text{ QIV, so } \theta = 300^\circ \quad (\text{real part positive, imaginary part negative})$$

$$\sqrt{5} \quad \text{first: } \frac{300^\circ}{2} = 150^\circ \quad \text{spacing: } \frac{360^\circ}{2} = 180^\circ, \text{ so } 150^\circ + 180^\circ = 330^\circ$$

$$\boxed{\sqrt{5} \text{cis } 150^\circ, \quad \sqrt{5} \text{cis } 330^\circ}$$

Take the root of r ($\sqrt{5}$), not half of it. The two square roots always point in opposite directions.

Find all roots. Answer in standard form.

$$13) x^3 - 64 = 0$$

Isolate x^3 , write the constant in trig form, take cube roots, then convert each root to standard form.

$$x^3 = 64 = 64 + 0i: \text{ positive real axis, so } r = 64, \theta = 0^\circ$$

$$\sqrt[3]{64} = 4 \quad \text{first: } \frac{0^\circ}{3} = 0^\circ \quad \text{spacing: } \frac{360^\circ}{3} = 120^\circ \quad \text{roots: } 4 \text{cis } 0^\circ, 4 \text{cis } 120^\circ, 4 \text{cis } 240^\circ$$

$$4 \text{cis } 0^\circ = 4 \quad 4 \text{cis } 120^\circ = 4 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = -2 + 2i\sqrt{3} \quad 4 \text{cis } 240^\circ = 4 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = -2 - 2i\sqrt{3}$$

$$\boxed{\left\{ 4, -2 + 2i\sqrt{3}, -2 - 2i\sqrt{3} \right\}}$$

Check another way: $x^3 - 64 = (x - 4)(x^2 + 4x + 16)$. The quadratic formula gives $x = \frac{-4 \pm \sqrt{16 - 64}}{2} = \frac{-4 \pm 4i\sqrt{3}}{2} = -2 \pm 2i\sqrt{3}$. A cubic always has 3 roots, so $x = 4$ alone is only a third of the answer.

Answers to Assignment (ID: 1)

- | | | |
|--|---|--|
| 1) 24 | 2) $6\sqrt{2} - 6\sqrt{6} + (-6\sqrt{6} - 6\sqrt{2}) \cdot i$ | 3) $1 + 12i$ |
| 4) i | 5) $\frac{3}{2} - 2i$ | 6) $-\frac{3}{10} + \frac{11}{10}i$ |
| 8) $-644 - 960i$ | 9) $-3479 + 1320i$ | 10) $\sqrt[5]{2} \left(\cos \frac{\pi}{15} + i \sin \frac{\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{13\pi}{15} + i \sin \frac{13\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{19\pi}{15} + i \sin \frac{19\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right)$ |
| | | 11) $\sqrt[5]{5}(\cos 24 + i \sin 24)$
$\sqrt[5]{5}(\cos 96 + i \sin 96)$
$\sqrt[5]{5}(\cos 168 + i \sin 168)$
$\sqrt[5]{5}(\cos 240 + i \sin 240)$
$\sqrt[5]{5}(\cos 312 + i \sin 312)$ |
| 12) $\sqrt{5}(\cos 150 + i \sin 150)$
$\sqrt{5}(\cos 330 + i \sin 330)$ | 13) $\{4, -2 + 2i\sqrt{3}, -2 - 2i\sqrt{3}\}$ | |

Every worked answer on pages 1–4 matches this key. #10 is given in radians here; the degree version ($12^\circ, 84^\circ, 156^\circ, 228^\circ, 300^\circ$) is the same set of roots.